

# Smart Meter Sensor Mounting Guide

This guide explains how to install non-invasive sensors on your household meters in Perth — covering electricity, gas, and water. These methods do not interfere with manual readings or metering accuracy.

## 1. Electricity Meter – Western Power EM1000 (Landis+Gyr)

- **Sensor Type:** Optical (photodiode/phototransistor) such as TCRT5000 or BPW40.
- **Signal:** 1 LED flash = 1 Wh (3600 flashes = 1 kWh).
- **Mounting:** Place sensor directly in front of the red LED marked “1 Wh/imp”. Use black tape or a 3D-printed shroud to block ambient light. Do not cover the screen.
- **Wiring:** Output to ESP32 GPIO via 10 kΩ pull-up resistor.
- **Notes:** Keep optical sensor dry and shielded from sunlight.

## 2. Gas Meter – ATCO Landis+Gyr G4 (Model 750)

- **Sensor Type:** Hall effect (A3144, US1881) or reed switch.
- **Signal:** 1 pulse = 0.01 m<sup>3</sup> gas.
- **Mounting:** Fix the sensor to the clear face above the red dial, near the digit “6”. Maintain ~2 mm gap from the glass. Do not block serial or reading digits.
- **Wiring:** Connect sensor Vcc (3.3–5 V), GND, and output to GPIO with 10 kΩ pull-up.
- **Notes:** Use silicone or tape, avoid adhesives that damage meter. Fully removable.

## 3. Water Meter – Elster/Itron Type (Brass, Underground Box)

- **Sensor Type:** Hall effect (A3144) or magnetic reed switch.
- **Signal:** 1 pulse = 1–10 L depending on dial marking.
- **Mounting:** Position sensor at edge of small red dial, not over the reading area. Attach using clear tape, magnet bracket, or removable silicone so manual reading remains visible.
- **Wiring:** Reed switch connects directly to GPIO and ground. Hall sensor requires 3.3–5 V, ground, and GPIO with pull-up.
- **Notes:** Seal sensor head only. Do not seal or obscure the full meter window.

## ESP32 Integration (for all sensors)

- Power: 5 V USB adapter (in sheltered or weatherproof box).
- Suggested GPIO assignments:
  - Electricity: GPIO12
  - Gas: GPIO13
  - Water: GPIO14
- Configure each in ESPHome or Arduino as a pulse counter.
- Send hourly data via Wi-Fi or MQTT.
- Mount ESP32 and wiring in IP65-rated box near meter cluster.